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Tracking and traceability in the meat processing industry: a solution

Tracking and
traceability:
a solution

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Keywords Food industry, Meat, Tracking

Abstract *The meat industry is seeking to establish reassurance on traceability and production techniques that may help to promote confidence in the integrity and origin of their products. This paper reviews research and development activities for traceability and trackability in the meat industry. Achievements of individuals and research groups developing tools and techniques to improve the production process in handling and cutting meat portions for end users are discussed. In addition software/hardware, logistics and technical requirements for tracking material in a production process are introduced. With the aid of such established tools and techniques coupled with novel proposals, the authors offer a practical solution for the development of a tracking and traceability system within the meat industry. The proposed prototype system is capable of identifying and handling a product and the information attached to it throughout the production process to retail packs.*

Introduction

In accordance with EC legislation consumers are seeking reassurance on traceability and production techniques that will re-establish confidence in the safety of meat products. The outbreak of livestock-related diseases and their possible transfer to humans have underlined the necessity for a reliable system to trace an individual meat product back to its animal of origin. To this end, all EU members were obliged to have a computerised cattle traceability system operational by the year 2000 (Furness, 1998). Such systems cover information such as: cattle passport database and movement data via electronic means taken up by auction markets, abattoirs, third party agents and organisations involved in movement of livestock. The systems developed and in use produce coherent labelling systems that reveal more accessible data ensuring the traceability of cattle starting from the farm through to abattoirs. Practitioners have proposed safety assurance frameworks that cover the entire supply chain. Specific standards cover the manufacture of feeding stuffs, transport, livestock markets, abattoirs, butchers, caterers, and retailers (Furness, 1998). The major activities pursued by research organisations in the meat industry may be outlined as:

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- production optimisation and monitoring systems;
- labelling and standardisation of tracking systems in the meat supply chain (*Meat Marketing Quarterly*, 1997);
- application of information technology and Internet-based systems.

According to this survey, substantial research has been dedicated to the development of automated tracking systems for the meat industry and successful implementation of cattle tracing systems from the farms to abattoirs. The complex disassembly of the carcass into muscle groups, meat cuts and trimmings during conventional meat processing batch boning represents a “black hole” with respect to traceability of the animal of origin although batch traceability has been successfully achieved. In recent years, researchers and practitioners have proposed tracking systems that enable producers to trace meat cuts from farm of origin to retail pack (Montanari and Coleman, 1995; Clarke, 1986, 1992; Pilz, 2000; Chanet and Eynard, 2000).

Traceability techniques proposed may be divided into two main categories. The first category is the usage of traceable equipment attached to the piece of meat such as readable tags (Montanari and Coleman, 1995), injectable transponders (Caja, 2000), ink jet marking and tattooing (Chanet and Eynard, 2000), and DNA tracing techniques (Meghen, 2000). The second category may be defined as traceability of material within the framework of material handling devices and automation (Pilz, 2000). The solution introduced in this paper falls into the second category. The introduction of a practical solution for total integration of a fully operational system that ensures 100 per cent traceability requires a comprehensive analysis of the current practice in abattoirs and boning halls. For successful implementation, modifications in the material handling devices, plant layout design, current processing practices, information flow and data handling procedures in addition to changes in operational culture seem inevitable.

The main challenges to be tackled in this system may be listed as:

- Design of an intelligent material-handling device that stores and transfers the information of a given piece of meat along its path. This device should be capable of diverting the pieces of meat from one path to another with minimal effort. It should be designed to ensure optimal ergonomics.
- Standardisation and automation of production process, enhanced tracking and labelling techniques, and compatible control systems are the main elements that may assist the system designer in solving complexities of carcass disassembly.
- Development of a standardised database system with uniform format along the path from farm to retail pack may aid in the integration of a proposed tracking system.

It is assumed that cattle are fully traceable from farm to the abattoir through the livestock supply chain and retail packs are traceable thereafter within the retail supply chain. The only part that is needed to complete the traceability in the supply chain is abattoir and boning hall activities. Once integrated into the system then the meat supply chain may be considered as fully traceable from “farm to plate”.

The following section provides a brief overview of the current technology available and introduces practical measures taken for development of a fully operational material handling system, which transfers the information associated with carcass from the abattoir through boning hall to retail packaging.

State-of-the-art

Automation and standardisation

It seems that some degree of standardisation and automation to minimise human errors is essential for tracking material within the abattoirs and boning halls. The standards could prevail in cutting techniques, material handling and process flow. The development and introduction of sophisticated automatic equipment has been impeded by the variation in carcass size and the non-homogeneous structure of the carcass further complicated by individual customer specifications (Clarke, 1986).

Longdell (1996) proposes some standard cutting and boning techniques as well as machines developed that may be used in standardisation and automation of the boning process. Clark (1986) also proposed some standard carcass splitting techniques for the breakdown of pork. Swain *et al.* (1990) express the need for a standard portioning scheme in poultry processing and offer design solutions for a robotic cell for optimal cutting. Despite these proposals and prototypes developed, the delicacy and the nature of the process of cutting and deboning individual muscle groups of red meat carcasses requires skilled human operators. Based on individual plant design and customer specifications the product and process may vary from operator to operator and from plant to plant. In addition, the standardisation and automation of meat processing faces natural resistance from the material itself (Clarke, 1992). Total quality management (TQM) techniques may help to alleviate the degree of difficulty in standardisation and automation in the meat processing industry. The food industry has been reluctant to join other manufacturing industries in adopting TQM (Mutri and Aly, 1989). Centralised purchasing with price emphasis, multiple vendors, dependence on nature, minimal training, and lack of standardised procedures are the main shortcomings of the industry (Mutri and Aly, 1989). However, such barriers may be addressed by taking measures like:

- granting business based on the quality of animal and ensuring that the resulting carcass meets some predefined specifications such as size, breed and age;

- usage of a limited number of suppliers from particular geographical areas to supply the specific breeds;
- establishment of stronger supplier customer relations to reduce the variability attributed to the raw material;
- effective training against documented operational procedures.

It is generally accepted that quality systems including operational structure, agreed procedures, capabilities and resources, and resulting activities provide confidence that products meet their quality requirements (Gerats, 1988). A degree of standardisation and automation within the framework provided by a documented quality system should therefore be achievable in cutting and deboning processes within boning hall activities.

Tracking systems

The main objective of a tracking system is to find best-fit technology to minimise the effort to capture timely and accurate information on the movement of material. Materials handling systems providing tracking have gained widespread popularity and have been incorporated into almost all sectors of industry (Billo and Bidanda, 1998).

The material tracking design framework (MTDF) introduced by Billo and Bidanda (1998) seems to be a comprehensive approach for designing and implementing a tracking and traceability system for meat processing. The framework may be used to identify the type and extent of data the system will carry, the functions and processes, communication, and the physical distribution and construction of the system.

There are some established tracing and tracking technologies that have been introduced to the meat industry by researchers and practitioners. These technologies include the following:

- *Bar codes*. The components of a bar code system consist of a scanning device, a portable terminal, and a label (Heard, 1991).
- *Microcircuit cards*. These cards can store data on the items being moved along the system by being attached to the item itself or the palette or the conveyor.
- *Radio frequency tags and transponders (RFTT)*. The most common application of non-contact RFTT microcircuit technology is monitoring the movement and location of pallets, containers, or vehicles as they move through the network to the ultimate receiver (Heard, 1991).
- *Radio frequency tags*. These are used to track movements of control assembly line operations, and provide for automatic identification. Their primary advantages are that they eliminate key entry, permit wireless operations, and reduce human intervention and error.
- *Voice recognition systems (VRS)*. Voice recognition technology allows users prompted by voice command to enter data on the status and

location of an item i.e. packages, pallets, containers, railcars, or vehicles directly into a computer simply by speaking into a microphone.

- *Biocoding*. Biocode technology has been developed so that individual batches of products can be identified.
- *Chemical markers*. These can be added at tiny concentrations (parts per billion or parts per million) to products such as stamps or tattoos in order to maintain and retrieve information during the process (Karimi and Lockhart, 1993).

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Process control

According to the production plan or customer requirements operational procedures may change significantly, making the control of process flow a challenging task. Despite such difficulties practitioners and researchers have proposed various techniques to control activities (Mutri and Aly, 1989; Wilkinson and Garner, 1992; MacDougall, 1988; Longdell, 1994; Laporte *et al.*, 1992; Shoup and Macchio, 1989).

MacDougall (1988) argues that, with increased understanding of the raw material variables and the manufacturing process, product quality should improve. This would be realised if developments in process technology were adequately controlled at all stages from animal selection through product manufacture to retail sale. The challenge facing the technologists is to control the raw material so that ultimate product uniformity is assured.

Trienekens and Trienekens (1993) offer a framework for information systems applications to deal with the increasing complexity in the food processing industry. It appears that the researchers and practitioners unanimously agree on the need for a standard process planning and control system in the food processing industry. However due to process constraints and the need for human intervention, acquiring such high levels of control and conformance to standard activities seem difficult. Nevertheless, with minimisation of natural variables and design of proper handling and processing equipment, coupled with control techniques, such obstacles are not insurmountable.

System integration

Established computing and hardware technologies may assist systems designers to develop and integrate fully operational tracking systems for the meat supply chain. The main objective here is to investigate a tracking system that is capable of handling and routing parts through various processes of cutting and boning. This tracking and traceability system may then be integrated into the livestock and retail supply chain to create a system that can relate individual prime cuts to the animal of origin.

Requirements for the development of a fully operational tracking and tracing system may be divided into hardware and software requirements. The hardware requirements deal with the basic physical auxiliary devices, material

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handling equipment and plant layout design, whilst the software requirements tackle control algorithms, labelling and coding techniques, read/write capabilities, software hardware interface, and systems integration.

Meat traceability: a solution

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The meat processing industry has established tracking systems capable of identifying origin of products through ear-tags. Ear-tags and other relevant information, e.g. breed, category, and grade, are transferred to the carcass at slaughter, but from slaughter onwards the animal tracking becomes increasingly difficult. Due to progressive disassembly the complexity of information and material-handling equipment increases. The tracking and tracing system proposed in this paper focuses on primal cuts whilst trimmings will also benefit from a less enhanced traceability.

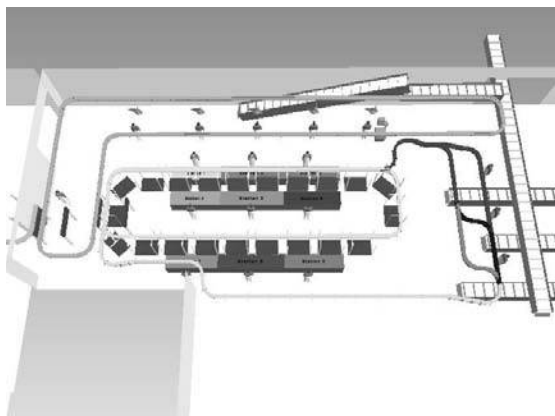
Conventional traceability is achieved through manual or computer assisted paper-based tracking systems. Typically these systems are based on the processor assigning each animal with a unique sequential production number, which corresponds to the animal's passport. This information is used in the proposed tracking system to transfer the data using combined radio frequency identification device (RFID) and barcode readers from the carcass to the Traceability Database and linking them to the plant's material handling system (MHS).

An appropriate MHS that restores and transfers animal information with the aid of advanced RFID and read/write equipment constitutes hardware requirements of the proposed system. The software part of the proposed tracking system fosters control algorithms, data flow and restoration, I/O terminals, and total integration of the tracing and tracking system.

Material handling system and radio frequency identification device

Automated material handling and RFID based tracking systems are currently available and common to many industries; these are mainly flatbed conveyors[1]. MHSs that are based on flatbed conveyor systems lend themselves to control complexities, extra storage equipment e.g. baskets and trays, and additional manual activities. The meat processing industry in the UK has been reluctant to adopt such systems, due to high costs of installation in the absence of any commercial benefit in a climate of financial decline through food scares, coupled with potential incompatibilities between systems, reliability issues and imposition of extra pressures on the operators.

Based on the prevailing constraints, research was conducted on a MHS that overcomes the limitations of current systems. For this reason a consortium of industrial companies and research groups, under a DEFRA funded Food LINK project named Meatracs, was formed to investigate and develop a novel MHS that automatically traces and tracks meat cuts within boning halls. Simulation techniques were conducted to analyse the present activities in boning halls and different material handling devices were tested to study the impact of using such systems within boning halls (see Figure 1).



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Figure 1.
Virtual model of the
Meatrac concept

Based on research and studies made on various techniques from economical and engineering aspects, an overhead conveyor system was considered to be a more cost-effective alternative, although overhead conveyors inherit their own limitations such as noise, complexity of drive systems and dirt traps along the path. However, the benefits in balance are utilisation of space and improved plant layout design, natural segmentation of meat cuts, maintaining traceability, cheaper solution to complex routings using junctions, minimal effort from operators in handling material, ease of interaction with software systems and remote control, and flexibility for different processing configurations.

Technical requirements for development of a low cost overhead conveyor system that is capable of automatically handling and routing large numbers of individual prime cuts to processing stations was the first stage of the project. A novel compression drive conveyor, in which a channel profile encapsulates a stream of solid spheres or beads, each being capable of supporting a carrier (tray) for single or multiple cuts of meat, was designed and manufactured. Figure 2 illustrates the beads in the channel profile of the proposed MHS.

THE INTELLIGENT CONVEYOR

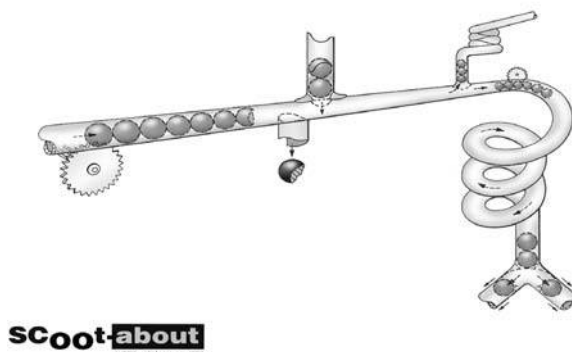


Figure 2.
Beads in channel profile
of a compression drive
conveyor system

The carriers (trays) are designed to transfer primal cuts along the designated track to the working stations where the operators receive the product for further processing. The trays will then be diverted and fixed using mechanical and electrical means to the working area of each station and act as a workbench for the allocated prime cut. The process of trimming is conducted on the tray and the bead is released back to the main track to be transferred to other stations. Use of a novel junction system to divert each bead to the working station clears the main path for other carriers to pass to other working stations. Figure 3 shows the concept of diversion of beads and trays and fixing them to the working area.

The carriers are designed to accommodate a microcircuit card (Read/write RF Smart label) with the unique code corresponding to animal information. This code will remain with the tray and the prime cut and its associated trim and fat through the process until the meat is removed from the tray and transferred to the corresponding packs. Figure 4 shows one possible scenario. Anti-collision techniques were developed for low cost RF tags, capable of reading information from individual moving tags.

Control systems and information handling

Development of control system protocols and algorithms, to allow the overhead conveyor to route primal meat cuts through the trimming and packing stations, automatically balancing material flows to demand requirements, was investigated. A thorough study of time and motion studies coupled with ergonomics of the plant layout design was conducted. System analysis based on industrial practice and line balancing helped the designers of the system to determine the throughput time and amount of work to be done at each station.

Control algorithms using PLC technology for remote control of conveyor system were designed to conform to the expected throughput of the system.

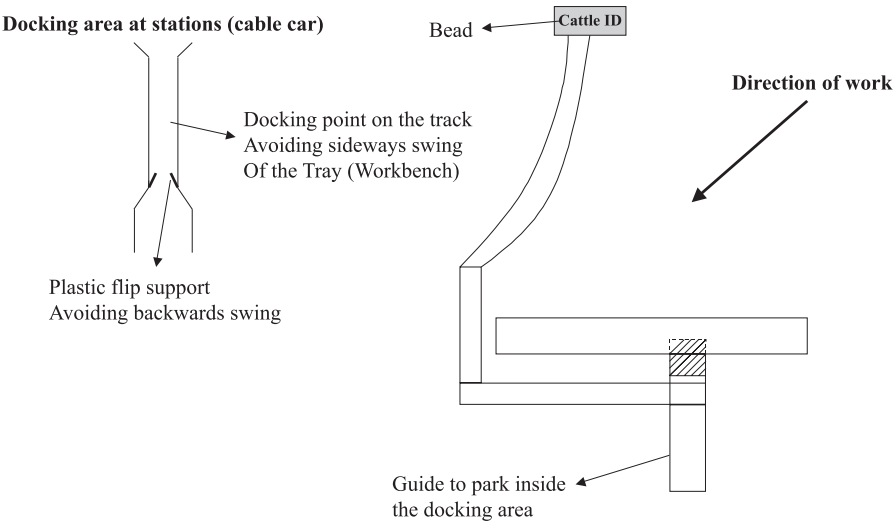
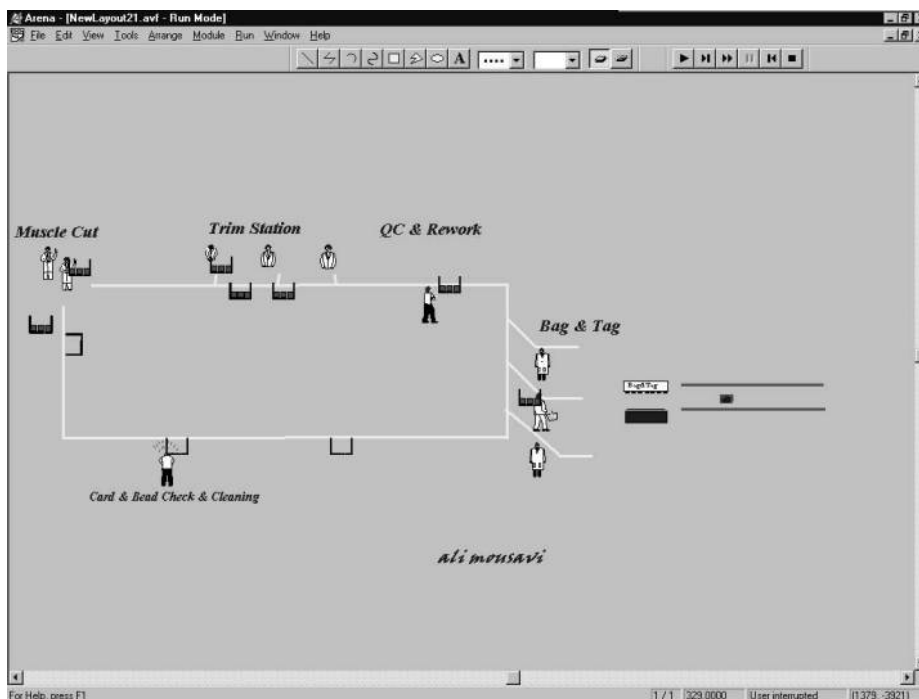


Figure 3.
Trays/junction/fixture



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Figure 4.
One possible scenario

Figure 5 illustrates the control system architecture of the system based on Rockwell's DeviceNet distributed Control System.

Integration devices were installed to ensure the flow of data from tags, tag readers, and DeviceNet to correlate the data with production planning database. Figure 6 shows the algorithm designed for software development.

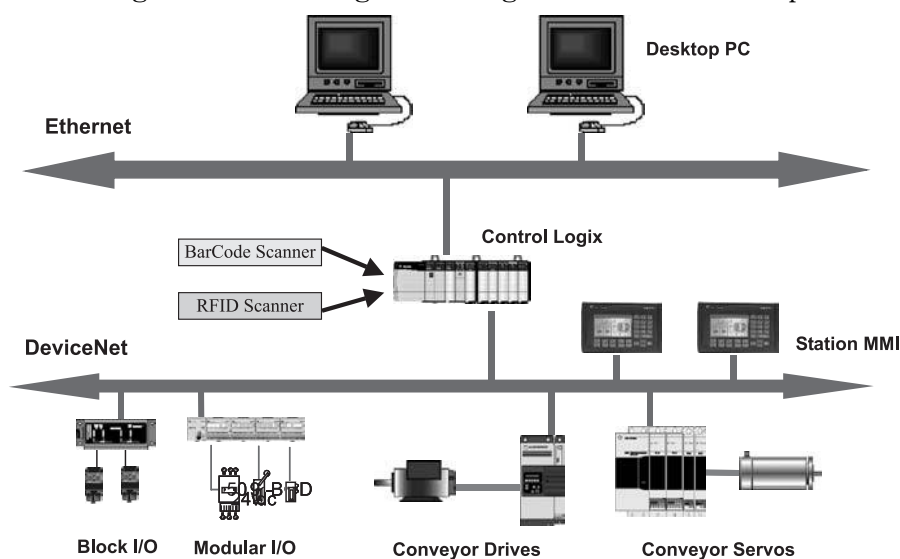


Figure 5.
Control system
architecture

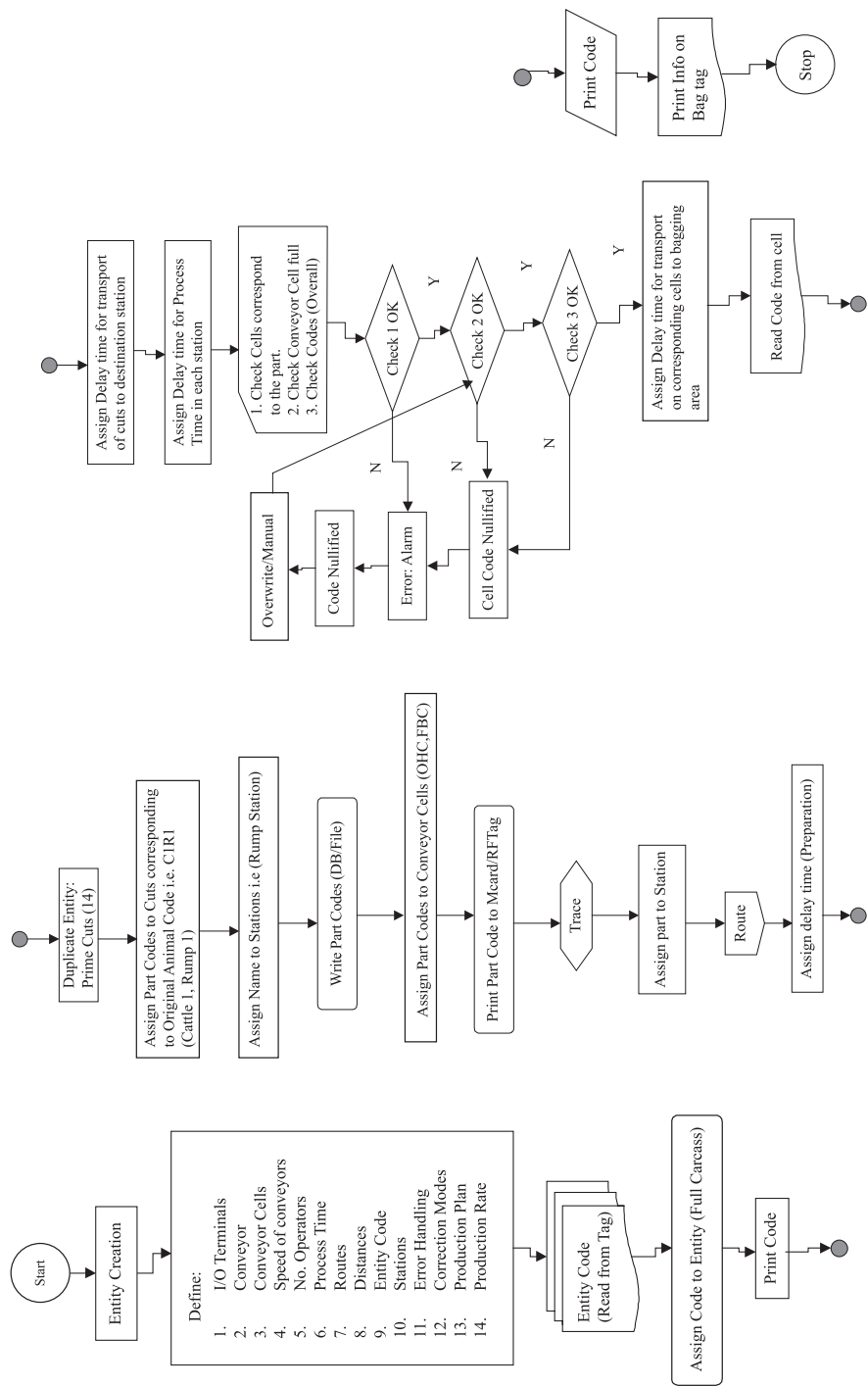


Figure 6.
Tracking system
algorithm

Each prime cut will have a unique identity that can be accessed along the system. This code will then be related to the corresponding carcass that holds a fully traceable passport to the farm of birth. Finally, in order to ensure the integrity of the system, generated codes may then be translated into commercial codes that are used by retailers such as URMIS (Uniform Retail Identity Standards).

To date, design of a working prototype of beads, tracks, drives and trays for the proposed overhead conveyor system has been made. A database system is designed for retrieval and storage of the obtained information. In addition interacting user-friendly front-end software has been developed for the user of the system to manage the queries from the database. The control system algorithms and the required control equipments have been designed. Analysis of the to-be system has been conducted using simulation techniques. What-if scenarios have been incorporated in the simulation analysis to review possible implementation problems. In future an Internet application of the software will be developed to enable users to retrieve information on the retail pack for full traceability to the animal of origin. Preliminary studies have shown a significant improvement in production process and yields and it is projected that the time scale for return of investment will be less than one year. The Meatrac project developed a full-scale pilot system sited at Pera to demonstrate proof of principle of the proposed solution. This demonstrator will be used to disseminate the benefits of this new system to the food industry and further afield.

Conclusions

The need for a fully operational and economically viable tracing system that is capable of tracing meat to the animal of origin is highlighted. In order to propose an integrated material handling system, techniques and methodologies proposed by contemporary researchers and practitioners are brought forward. This paper highlights the importance of boning hall activities in the process of tracking meat products within the total meat supply chain. A practical system is proposed, that focuses on traceability of individual prime cuts in the boning hall to the animal of origin. The information gathered is then stored in a database and can be accessed and translated into standard commercial formats.

This paper presents a novel approach in designing an intelligent overhead conveyor system that is capable of ensuring maximum traceability. In addition it opens a window to a novel approach in software/hardware interface for control systems that track and trace product along meat production process. It will help in improving the yield and quality of product by ensuring best practice in meat processing and material handling.

Such a system may further be expanded as an Internet-based application where consumers may directly interface with meat production plants and investigate the origins of their purchased product.

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Note

1. Axman bi-directional, SFK Denmark, Stork of Holland and Siemens flatbed conveyor systems.

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